

Reading, Not Reasoning: A Causal Faithfulness Audit of Chain-of-Thought in Distilled Chart-and-Table VLMs

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Abstract

Many multimodal systems compress agentic visual reasoning into a single tool-free VLM forward pass via chain-of-thought supervised fine-tuning. We ask what such distillation actually transfers. Our causal claims concern the *emitted written chain*, not hidden-state or latent computation that never surfaces as text. Across chart, table, natural-image counting, and rendered financial QA, CoT SFT improves accuracy, often by large margins, yet a causal faithfulness audit shows that the emitted written chain is not load-bearing: corrupting an on-path numeric intermediate does not produce a reliable excess flip over no-target controls, including a targeted same-shape local numeric control, and the model almost never follows the injected value, instead snapping back to the answer readable from the original evidence. The information source behind each gain is task-dependent: the answer remains recoverable from the visible chart or table and the model snaps back to it, is reconstructed from redundant enumerations in the rationale, or is copied from the chain’s own conclusion. Even a shortcut-removing curriculum with gold-program-verified operands raises accuracy substantially without installing load-bearing intermediate computation. Accuracy-only internalization claims conflate reading with reasoning; causal probes over emitted rationales can separate the two.

1 Introduction

Vision-language models increasingly answer chart, table, and figure questions by writing out a chain of intermediate steps, and the strongest systems wrap several such passes inside an agentic loop of iterative frame retrieval, self-reflection, critic-orchestrated re-reading, and visual tool use. These procedures raise accuracy, but the tools and repeated forward passes they depend on are costly to run at inference time. A common response is to compress the loop: an expert trajectory is converted into a written rationale or a latent trace, and a single VLM is fine-tuned to reproduce the answer in one forward pass (Adhikari and Lapata, 2026; Carbune et al., 2024; Hu et al., 2024; Tao et al., 2026; Wei et al., 2026).

When such distillation lifts accuracy, the gain is usually read as evidence that the student has internalized the reasoning the trajectory performed. That reading skips a prior question. A written chain that accompanies a correct answer need not be the chain that produced it: the emitted intermediate steps may carry the computation, or they may record, paraphrase, or simply reveal an answer the model recovered some other way. On chart and table tasks the visual evidence stays fixed in the input, so an answer the model could read directly is never far from one it could compute, and accuracy alone cannot tell the two apart.

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We test which of these holds. We assess causal faithfulness by corrupting an emitted numeric intermediate and asking whether the final answer tracks the edited value, comparing against no-target controls that range from harsh order disruption to a gentler same-shape local numeric perturbation, together with a targeted semantic counterfactual diagnostic (Lanham et al., 2023). The claim we can support concerns the emitted written chain: the probes intervene on text the model actually wrote, and do not rule out hidden-state or latent computation that never surfaces in the rationale.

The paper makes four connected points. First, CoT SFT does improve accuracy on chart, table, natural-image counting, and rendered financial QA, so the audit interrogates a real accuracy signal rather than a null baseline. Second, that signal and causal faithfulness come apart: a probe battery across the four task types finds the emitted rationale is not load-bearing even where the accuracy gain is large and statistically robust. Third, where the answer actually comes from is task-dependent, and the controls are built to tell the cases apart: the model recovers the answer from still-visible evidence, exploits a redundant enumeration elsewhere in the rationale, or copies the chain’s own conclusion. Finally, a pre-specified curriculum that removes shortcuts and verifies operands against a gold program still raises accuracy without installing a load-bearing intermediate, which rules out the simplest training-data explanation for the null.

2 Related Work

Internalizing visual reasoning. Visual Program Distillation (Hu et al., 2024), chart-specific rationale transfer (Carbune et al., 2024; Chen et al., 2025), latent trajectory learning (Adhikari and Lapata, 2026; Li et al., 2026), and region-to-image or local-cue distillation (Tao et al., 2026; Wei et al., 2026) all aim to move work that previously required tools, crops, or expert traces into a cheaper model or inference path. These papers mainly validate the transfer by answer accuracy, sometimes with attention or qualitative trace analysis. Our question is narrower: after a no-tool VLM emits a discrete rationale, is the emitted intermediate value an answer-carrying state, or is it an artifact of a successful read?

Chain-of-thought and its faithfulness. Chain-of-thought prompting (Wei et al., 2022), its zero-shot trigger (Kojima et al., 2022), and self-consistency decoding (Wang et al., 2023) made written intermediate steps a default interface for reasoning, which in turn raised the question of whether those steps are faithful to the computation that produces the answer. A broader explanation-evaluation literature had already cautioned that a plausible rationale need not be the true cause of a prediction (Jacovi and Goldberg, 2020), that rationale datasets and metrics measure heterogeneous properties (Wiegrefe and Marasović, 2021), and that rationalized models must be benchmarked against the evidence they actually use (DeYoung et al., 2020).

Within text LLMs the question has produced more specific results. Rationale scaffolds can be made more explicitly faithful (Lyu et al., 2023), plausible rationales can omit true decision drivers (Turpin et al., 2023), larger models can produce less answer-dependent CoT (Lanham et al., 2023), and filler tokens can support hidden computation independent of human-readable steps (Pfau et al., 2024). Two critiques bear directly on our design: unlearning or deletion can measure contextual robustness rather than load-bearing computation (Tutek et al., 2025), and multiple-choice option bias can make some faithfulness metrics look like disguised accuracy (Bentham et al., 2024), which is part of why our substrates are free-response numeric QA.

In multimodal settings the audited object usually differs from ours. Visual-thinking agents may ignore visual evidence (Liu et al., 2025b), CodeV perturbs tool-output images rather than internalized CoT tokens (Hou et al., 2026), and latent imagination work studies continuous intermediate states

(Li et al., 2026).

Chart/table QA and tool/program distillation. Chart and table QA are useful because the same task mixes visual extraction and numerical reasoning. Benchmarks from synthetic figures (Kafle et al., 2018; Kahou et al., 2018; Methani et al., 2020) to realistic charts (Masry et al., 2022; Tang et al., 2025; Wang et al., 2024), semi-structured math tables (Lu et al., 2023), and hybrid financial tables (Chen et al., 2021; Zhu et al., 2021), together with chart-noise stress tests (Shin et al., 2025) and perception-bottleneck analyses (Liu et al., 2025a), all point to a persistent ambiguity: an accuracy improvement can come from reading values better, computing with them better, or both. A parallel line tries to install the computation explicitly, either by emitting executable programs at inference (Chen et al., 2023; Gao et al., 2023), learning to call tools (Schick et al., 2023), or distilling tool traces and step-by-step rationales into a cheaper model (Hsieh et al., 2023; Hu et al., 2024). Our experiments exploit the reading/computing ambiguity instead of hiding it: by masking images, falsifying on-path operands, and comparing chart/table with natural-image counting, we ask which representation the answer actually reads from at the final step.

Positioning. Table 1 places this work against neighboring lines along four axes: the object audited, the intervention used, the model setting, and the conclusion the design licenses. The novelty is not the CORRUPT intervention, which is a standard counterfactual edit (Lanham et al., 2023). It is that the audited object is the discrete CoT of a *distilled, single-forward* VLM whose visual evidence is fixed in the input, evaluated with controls that separate re-reading, redundancy, and conclusion copying. That combination lets us ask not just whether the chain is faithful, but which fixed source the answer reads from when it is not.

3 Method: Causal Faithfulness Audit

This section defines the audit, not just the datasets, models, and metrics it runs on. We specify how a targeted counterfactual edit (CORRUPT), a battery of no-target controls (SHUFFLE, LOCAL-NUM, and others), and a snap/follow/other readout together decide whether an emitted written chain is *load-bearing*: whether the final answer causally tracks an edited intermediate, or instead reads from a source still available at answer time.

Definitions. We fix three terms used throughout. A *load-bearing written chain* is an emitted numeric intermediate whose value causally determines the final answer: under a counterfactual edit to that token, a load-bearing chain forces the answer to track the edited value. Operationally we require *two* signatures together, not either alone: (i) the edit produces an excess flip over a no-target control ($F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{CONTROL}} > 0$), and (ii) the new answer *follows* the injected value rather than snapping back to the original truth or derailing into an unrelated answer. A stricter intervention that raises the raw flip rate but moves the extra mass into neither-snap-nor-follow “other” responses therefore does not by itself establish a load-bearing chain. *Reading* is the complementary case in which the final answer is recovered from a source still available at answer time, such as visible pixels, a redundant enumeration elsewhere in the rationale, or the chain’s own conclusion line, so that editing an upstream intermediate leaves the answer unchanged. *Latent reasoning* is any computation that does not surface as an edited written token, including arithmetic that the model may perform while re-reading the image. Our probes intervene only on emitted written tokens; they can show that a written intermediate is not load-bearing, but they cannot certify the absence of latent reasoning.

Table 1: Field positioning. Where a causal CoT audit of distilled chart/table VLMs sits relative to neighboring lines of work. “Object audited” is the representation perturbed; “conclusion licensed” is what the design can actually claim.

Line of work	Object audited	Intervention	Model setting	Conclusion licensed
Text CoT faithfulness (Bentham et al., 2024; Lanham et al., 2023; Pfau et al., 2024; Turpin et al., 2023; Tutek et al., 2025)	Written CoT tokens	Truncate / paraphrase / corrupt / unlearn	Prompted text LLM	Whether text CoT is answer-bearing
Rationale / explanation eval. (DeYoung et al., 2020; Jacovi and Goldberg, 2020; Wiegrefe and Marasović, 2021)	Highlighted rationale / evidence	Erasure, comprehensiveness, sufficiency	Classifier + rationale	Whether a rationale matches used evidence
Visual-agent faithfulness (Hou et al., 2026; Liu et al., 2025b)	Tool-output images / agent traces	Perturb tool outputs / RL reward	Tool-using VLM agent	Whether agents use visual tool results
Tool / program distillation (Chen et al., 2023; Gao et al., 2023; Hsieh et al., 2023; Hu et al., 2024; Schick et al., 2023)	Program / tool trace	Train-time supervision; run program	Distilled / program-aided model	That accuracy transfers (faithfulness untested)
Latent-state reasoning (Adhikari and Lapata, 2026; Li et al., 2026)	Continuous latent states	Probe / edit embeddings	Latent-trace VLM	Whether latent states carry reasoning
Chart/table QA benchmarks (Chen et al., 2021; Kafle et al., 2018; Lu et al., 2023; Masry et al., 2022; Methani et al., 2020)	Final answer only	None (accuracy scoring)	Any VLM/LLM	Aggregate accuracy on visual+numeric QA
This work	Emitted CoT of distilled single-forward VLM	Targeted numeric corrupt + no-target controls + mask	Distilled VLM, fixed visual input	Which fixed source the answer reads from

“Reading, not reasoning” is therefore a claim about the emitted chain, not about the model’s internal state.

Tasks. *ChartQA* (Masry et al., 2022) (400 held-out open numeric questions, image-disjoint from SFT train images), *TabMWP* (Lu et al., 2023) (400 rendered-table, free-response numeric cases), TallyQA-complex natural-image counting (400 open integer questions), and FinQA rendered tables for the pre-specified curriculum probe. All four tasks fix the visual evidence in the input, making the causal question well-posed: does the written chain carry the computation that produces any SFT accuracy gain?

Models. The main SFT and Qwen-family faithfulness cells use Qwen3-VL 8B and 32B (dense) (Bai et al., 2025). For cross-family external validity, we also run the same ChartQA present-image probe on InternVL3.5-8B (Wang et al., 2025), a non-Qwen VLM served through an OpenAI-compatible local endpoint. All models are served locally.

SFT pipeline. For ChartQA and TabMWP, a Qwen3-VL-32B teacher generates step-by-step rationales on training images; we keep only traces whose final answer matches the gold answer under the numeric grader below. This yields 150 ChartQA traces and 160 TabMWP traces. The 8B students are LoRA-SFT models with the vision tower frozen, rank 16, $\alpha=32$, dropout .05, bf16, learning rate 10^{-4} , per-device batch size 1, gradient accumulation 8, cosine schedule, and prompt tokens masked to -100 so supervision applies only to the assistant rationale and ANSWER line. The ChartQA/TabMWP 8B runs use two epochs, corresponding to 38 and 40 optimizer updates. The 32B students use the same LoRA target modules under NF4 QLoRA and run five epochs with greedy held-out evaluation after each epoch; the selected checkpoints are epoch 1 for ChartQA and epoch 4 for TabMWP. Dense 8B controls use full-SFT checkpoints with the vision tower, embeddings, and first three language layers frozen, leaving about 79.7% of parameters trainable. The FinQA curriculum arms use the same LoRA recipe for three epochs: 180/180 gold-program strict/vanilla traces, and 173/173 answer-conditioned text-teacher strict/vanilla traces. All answer generation and probe continuations use greedy decoding (`do_sample=False`; answer max-new 320–384 tokens, continuation max-new 64 tokens).

Probe construction. For each task/model cell, we first ask the evaluated model to solve the original visual question with the image/table visible and the prompt **Solve step by step, end with 'ANSWER: <final>'**. The text before the first **ANSWER:** is the emitted CoT. A case is probe-eligible only if this CoT is nonempty and the extracted final answer is correct. Empty rationales, abnormal outputs, failed final-answer parses, and wrong final answers are not edited; they re-enter the analysis only through the selection-flow and sensitivity bounds below. Eligible CoTs with no corruptible numeric token remain in the flow but are excluded from the CORRUPT denominator and treated as corrupt-unusable in sensitivity bounds. The generic chart/table battery then edits that model-owned CoT with a fixed edit seed `Random(0)`: CORRUPT samples one numeric token from the CoT and replaces value v by $2v + 7$ (integers remain integers; decimals are rendered to one decimal), while SHUFFLE shuffles sentence order without injecting a target value. Additional controls paraphrase the CoT with an exact number-multiset guard, replace tokens with length-matched filler, truncate early sentences, or delete the last number-bearing steps. A gentler LOCAL-NUM control excludes the CORRUPT span, replaces a different numeric token with a same-shape value, preserves sentence order and local syntax, and avoids the gold/base/injected values when possible. A stronger whole-prefix SEMANTIC-CF rewrite is reported in the supplementary material: it sends the probe-eligible CoT prefix to a configured LLM rewrite endpoint, changes non-final numeric reasoning with locally coherent dependent arithmetic, and preserves the protected final conclusion segment. The FinQA targeted probe is stricter: it uses the gold program to choose a table operand that is non-constant, appears in the student’s CoT, and whose replacement by $2v + 7$ changes the gold-program final answer; it then tests both operand-only corruption and a consistently propagated corrupted chain. Each edited CoT is force-continued with the prompt “Reasoning so far: ... Given ONLY the reasoning above, state the final answer now as **ANSWER: <final>**”. The masked condition drops the image/table from this force-continue call while keeping the same edited text.

Grading and readout. Open numeric answers are graded in the task’s original answer units. We strip commas and percent signs from the gold answer, extract all numeric strings from the prediction, and mark a match if any predicted number is within $0.05|g| + 10^{-6}$ of gold value g ; nonnumeric gold answers fall back to normalized substring matching. Rounding and redundant units are therefore tolerated only through the numeric value; symbolic equivalents that do not surface a matching number, empty outputs, and parse failures are incorrect. “Flip” means the forced final answer no

Original CoT: read value a , read value b , compute $a - b = c$, answer c . CORRUPT: read value a , read value b , compute $a - b = \tilde{c}$, answer ? SHUFFLE: same rationale tokens, shuffled order, answer ? LOCAL-NUM: same rationale order, different local number perturbed, answer ? Readout: snap to true c / follow injected $\tilde{c}$ / other.

Figure 1: Probe schematic. CORRUPT tests whether a falsified written intermediate is load-bearing; SHUFFLE is a harsh no-target order control, while LOCAL-NUM is a gentler same-shape local numeric control.

longer matches the unedited model answer. For CORRUPT and FinQA targeted probes, the three-way readout is applied in priority order: snap if the answer matches the gold answer, follow if it does not snap but matches the injected wrong value (or injected gold-program final), and other otherwise. Boundary examples under 5% tolerance are: gold 100, injected 207, answer “about 96” $\rightarrow$ snap; answer “207” $\rightarrow$ follow; answer “200” $\rightarrow$ follow; answer containing both 100 and 207 $\rightarrow$ snap by precedence; and a SHUFFLE answer has no injected target, so off-gold responses are counted as other rather than follow.

Statistics. McNemar exact tests for paired SFT deltas, using only discordant pairs; Holm correction across the four headline ChartQA/TabMWP cells leaves all tests below .01. At $n=400$, minimum detectable net is about 8–9 points, and the observed paired SFT gains have 66 and 46 discordant pairs at 8B/32B. For the faithfulness probe, the first load-bearing signature is the one-sided test $\text{flip}_{\text{CORRUPT}} > \text{flip}_{\text{SHUFFLE}}$, Holm-corrected across the eight Qwen chart/table cells, with bootstrap intervals on F ; the second signature is the follow rate, reported with Wilson intervals on snap/follow (§5). Both must hold for a chain to count as load-bearing (§3).

Robustness controls. We do not claim the CORRUPT intervention itself is new; it is useful only as part of a controlled battery. To address the “contextual faithfulness” concern (Tutak et al., 2025), we compare CORRUPT against SHUFFLE, paraphrase, filler-token, deletion, truncation, image-mask, a targeted LOCAL-NUM replication, and a snap/follow/other readout that asks whether the final answer follows the injected value. To address disguised-accuracy and option-bias concerns (Bentham et al., 2024), the main faithfulness substrates are free-response numeric QA rather than multiple choice. To address the concern that force-continue is a two-pass re-prompt rather than an on-stream causal intervention, we additionally validate with an IN-PLACE paradigm that corrupts a token inside the model’s own greedy generation stream and lets it continue (Result II, finding 7). To avoid reporting a probe artifact as mechanism, we treat a positive corruption flip as insufficient by itself; the headline evidence is convergence across controls and regimes.

4 Result I: CoT SFT Improves Accuracy

On the chart and table tasks, we run the actual distillation. A Qwen3-VL teacher generates step-by-step rationales; answer-consistency filtering keeps high-quality traces; LoRA/QLoRA SFT trains a single-forward student. All reported evaluation sets are image-disjoint from training and use open numeric grading. These gains are best read as improved single-pass readout unless the causal probe shows that the emitted chain itself carries the answer.

On ChartQA ($n=400$), accuracy rises from .713 to .778 at 8B (net +6.5 points, CI [+2.5, +10.5], McNemar $p=0.0021$) and from .725 to .795 at 32B (net +7.0 points, CI [+3.8, +10.3], $p=6.9 \times 10^{-5}$). The original 32B $n=60$ pilot had only four discordant pairs, whereas the expanded evaluation has 37

gains against 9 losses. On TabMWP the gains are larger still: .865 to .963 at 8B (net +9.8 points, $p=2.9\times 10^{-9}$) and .835 to .973 at 32B (net +13.8 points, $p=2.1\times 10^{-12}$).

The paper is not, then, a null-result critique of SFT. The accuracy gains are real, paired, and adequately powered on static chart/table substrates. The remaining question is what mechanism those gains bought.

Dense 8B control. To rule out a low-rank-capacity explanation, we repeat the 8B arms with dense/full-SFT controls (vision tower, embeddings, and the first three language layers frozen; about 79.7% trainable parameters for ChartQA/TabMWP). The accuracy deltas remain positive: ChartQA .713 $\rightarrow$.763 (net +5.0 points, 39 gains/19 losses, $p=0.0126$) and TabMWP .865 $\rightarrow$.968 (net +10.3 points, 42 gains/1 loss, $p=1.1\times 10^{-9}$). Available dense probes keep the LoRA signature: ChartQA’s completed six-intervention battery has present-image CORRUPT/SHUFFLE = .100/.259, snap .917 and follow .030; TabMWP’s completed core battery has present-table CORRUPT/SHUFFLE = .049/.161, snap .971 and follow .016; FinQA dense targeted arms keep operand-follow near zero while consistent-conclusion follow remains .947–.982. The dense control thus rules out a low-rank-capacity explanation for both the accuracy gain and the non-load-bearing signature.

5 Result II: The Causal Probe Shows the Written Chain Is Not Load-Bearing

For every correctly answered case with an emitted CoT, we intervene on the rationale and force-continue to a final answer. If a numeric intermediate carries the computation, CORRUPT should flip more often than no-target controls, and the final answer should follow the injected value rather than snap back to the original truth. Table 2 reports the high-power Qwen-family ChartQA and TabMWP CORRUPT-vs.-SHUFFLE cells with uncertainty; this family treats SHUFFLE as a deliberately harsh order/format control, not as the only clean control. The confirmatory family is the eight Qwen chart/table CORRUPT-vs.-SHUFFLE cells; the load-bearing alternative is one-sided, $\text{flip}_{\text{CORRUPT}} > \text{flip}_{\text{SHUFFLE}}$, with Holm correction across that family. Table 3 separately reports the non-Qwen cross-family replication and is not included in that Holm family. Snap/follow intervals are Wilson 95% intervals. The legacy headline battery stores per-case CORRUPT answers and aggregate SHUFFLE margins, so F intervals use an aggregate binomial bootstrap and the test is an exact randomization conditional on the recorded margins; newer battery files with `details[]`.answers use paired case bootstrap and paired exact sign-randomization under the same script. The LOCAL-NUM replication is reported in the text below; the stronger whole-prefix SEMANTIC-CF diagnostic is reported in the supplementary material and `faithfulness_stats.md`.

The probe returns the same verdict across every cell, and the agreement between ChartQA and TabMWP is the first sign of it. In each present-image/table cell CORRUPT $\leq$ SHUFFLE within the 95% interval, none of the Holm-corrected load-bearing tests comes close to significance, and the model usually snaps back to the true value rather than following the injected one. TabMWP is the cleanest case: despite large SFT gains, follow is .008 [95% CI .003,.023] at 8B and .015 [.007,.033] at 32B. The signature is not specific to the Qwen family. InternVL3.5-8B answers 302/400 ChartQA cases correctly with an emitted CoT and behaves the same way, with CORRUPT/SHUFFLE = .070/.096, $F = -.026$ [95% CI $-.053, -.003$], snap .990, and follow 0/302 [95% CI .000,.013]. This does not show that Qwen-style SFT transfers to InternVL; it answers the narrower concern that low follow and high snap might be a Qwen3-VL-specific probe behavior.

The decoupling holds on exactly the cases the gain is made of. Joining the probe-kept cases to the paired SFT evaluation labels (Table 4), present-condition follow is zero on every set of cases newly

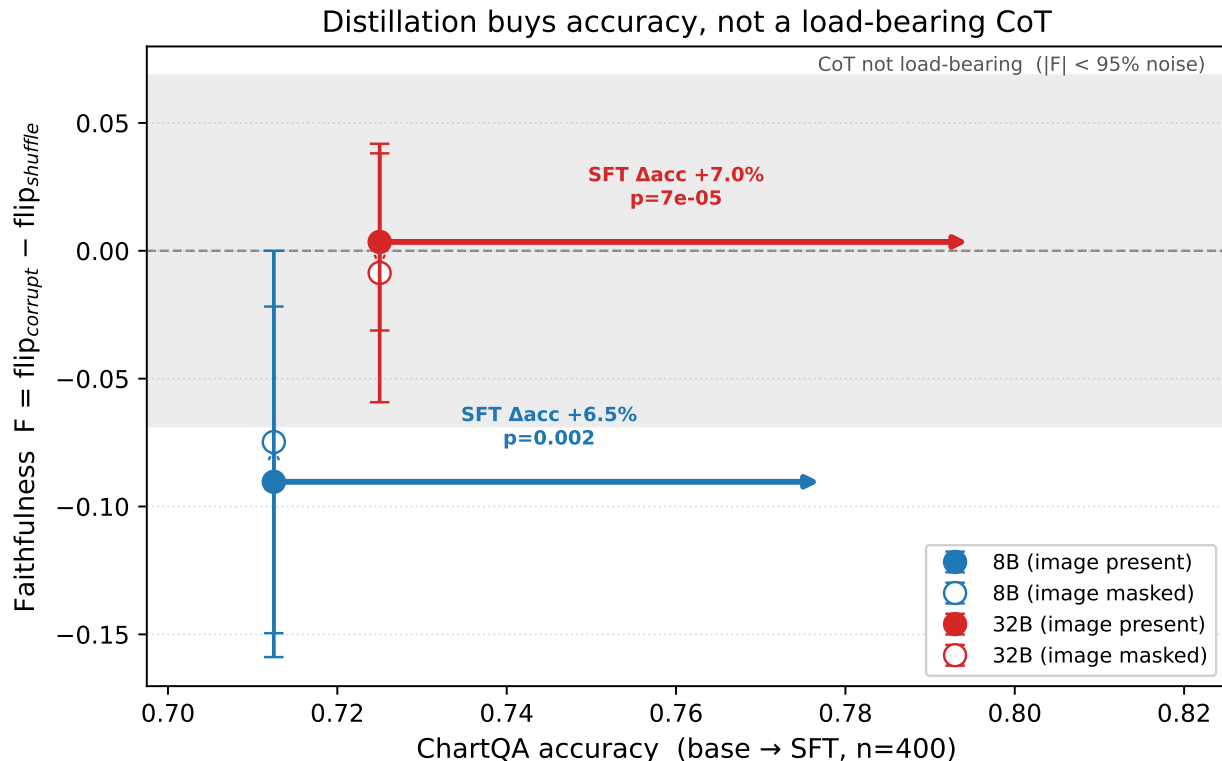


Figure 2: Accuracy $\perp$ written-chain faithfulness (chart/table = pixels-source regime). CoT distillation lifts ChartQA accuracy by +6.5–7.0% (McNemar $p \leq .002$; horizontal arrows), yet the causal CoT metric stays on the $F \approx 0$ band ($F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{SHUFFLE}}$): corrupting a numeric intermediate flips the answer no more than shuffling the CoT. This is a conservative harsh-control criterion: SHUFFLE disrupts order without injecting any target value, so $F \leq 0$ means the falsified intermediate contributes no extra flip beyond order/format noise. Flip rates here are from the two-pass force-continue probe; a stricter generation-time probe (finding 7) raises the raw flip rate without making the answer follow the injected value, so the load-bearing conclusion is unchanged. A targeted ChartQA 8B LOCAL-NUM replication below checks the same claim under a gentler no-target perturbation. Hollow markers show image-masked controls. F is measured on the same SFT student used for the corresponding probe; the tabular intervals and corrected tests in Table 2 are regenerated by `scripts/faithfulness_stats.py`.

solved by SFT: ChartQA 8B 0/39 [95% CI .000,.090], ChartQA 32B 0/37 [.000,.094], TabMWP 8B 0/40 [.000,.088], and TabMWP 32B 0/57 [.000,.063]. FinQA is not reported as a matched gain subset, because its targeted battery is keyed by gold-program operand eligibility rather than by base-wrong $\rightarrow$ SFT-right eval partitions, but its operand-follow intervals are equally tight: 0/172 at 8B [0,.022] and 0/175 at 32B [0,.021]. The answers SFT newly gets right are not being computed from the falsified written arithmetic.

Selection into the probe does not manufacture this null. The estimand is conditional on an emitted, correct written rationale by construction, since an incorrect or missing rationale cannot be asked whether its intermediate value was load-bearing for a correct answer, but the selection flow is still visible. TabMWP keeps almost every SFT-correct case (385/385 at 8B, 389/389 at 32B) and its gain strata have zero present-condition follows; ChartQA is noisier, with 10 SFT-correct answer-eval cases missing from probe eligibility at 8B and 20 probe-eligible cases that were not SFT-correct in the separate answer-eval pass, which we report rather than silently drop. A conservative bound that assigns every unobserved raw case to follow gives a loose all-raw upper bound of .223 for ChartQA

Table 2: Causal CoT-intermediate probe on chart/table SFT cells. Flow is raw $n=400$ to probe-eligible correct+CoT cases, with CORRUPT/SHUFFLE denominators in parentheses. $F = \text{flip}_{\text{CORRUPT}} - \text{flip}_{\text{SHUFFLE}}$; p_H is Holm-adjusted for the one-sided load-bearing test. Snap/follow are rates after CORRUPT with Wilson 95% CIs.

Dataset/model	Cond.	raw→probe	CORRUPT/SHUFFLE	F [95% CI]	p_H	snap [95% CI]	follow [95% CI]
ChartQA 8B	present	400→321 (321/321)	.212/.302	−.090 [−.159, −.022]	1.000	.816 [.770, .855]	.031 [.017, .056]
ChartQA 8B	masked	400→321 (321/321)	.333/.408	−.075 [−.150, +.000]	1.000	.710 [.658, .757]	.100 [.072, .137]
ChartQA 32B	present	400→318 (316/318)	.051/.047	+ .003 [−.031, +.038]	1.000	.981 [.959, .991]	.016 [.007, .036]
ChartQA 32B	masked	400→318 (316/318)	.120/.129	−.009 [−.059, +.042]	1.000	.908 [.871, .935]	.057 [.036, .088]
TabMWP 8B	present	400→385 (385/385)	.034/.145	−.112 [−.151, −.073]	1.000	.971 [.950, .984]	.008 [.003, .023]
TabMWP 8B	masked	400→385 (385/385)	.278/.299	−.021 [−.086, +.044]	1.000	.722 [.675, .764]	.088 [.064, .121]
TabMWP 32B	present	400→389 (389/389)	.033/.072	−.039 [−.069, −.008]	1.000	.969 [.947, .982]	.015 [.007, .033]
TabMWP 32B	masked	400→389 (389/389)	.075/.103	−.028 [−.069, +.010]	1.000	.928 [.898, .950]	.039 [.024, .063]

Table 3: Cross-family replication. InternVL3.5-8B is a non-Qwen VLM evaluated on the same ChartQA present-image probe protocol. This row is external-validity evidence and is reported separately from the Qwen-family SFT cells in Table 2.

Model family	raw→probe	base acc	CORRUPT/SHUFFLE	F [95% CI]	p	snap [95% CI]	follow [95% CI]
InternVL3.5-8B	400→302 (302/302)	302/400 (.755)	.070/.096	−.026 [−.053, −.003]	.994	.990 [.971, .997]	.000 [.000, .013]

present cells, yet that adversarial figure is contradicted by the directly observed SFT-correct subset: even if every SFT-correct ChartQA 8B case missing from the probe had followed the injected value, SFT-correct present-condition follow would be at most $19/311=.061$, ChartQA 32B at most $7/318=.022$, and TabMWP below .016. The conclusion does not rely on keeping only stable-correct cases.

Masking the image turns the probe into a diagnosis of where the answer comes from. On tables, removing the image raises flip and “other” errors, most of all at 8B, while follow stays small. That contrast is the key: were the model obeying the corrupted chain, dropping the image should leave follow unchanged or higher, whereas follow instead stays low while “other” rises, so the model is neither faithfully using the bad chain nor adopting the falsified value but losing the source it actually relied on. We therefore state the visible-input mechanism conservatively. The answer remains recoverable from the visible chart or table and the model snaps back to it; because we do not run targeted cell, bar, or region editing, we do not claim a literal re-reading saccade, only recoverability of the answer from fixed visual evidence rather than from the obeyed written intermediate.

The remaining interventions reinforce this reading rather than carry it, with one caveat worth stating. Paraphrase behaves like CORRUPT (for example, 32B ChartQA present .053 vs. .051), while filler, truncation, deletion, and SHUFFLE mainly test format and token-budget sensitivity. Because SHUFFLE injects no target value, its paper-facing readout is flip/snap/other rather than follow, which confirms it as a harsh control rather than an injected-value test, with per-cell SHUFFLE accuracy-after rates in the supplementary material. A gentler LOCAL-NUM replication on ChartQA 8B gives the same qualitative null under paired analysis: CORRUPT/LOCAL-NUM = .235/.224 on the 294 cases with both continuations, $F_{\text{local}} = +.010$ [95% CI −.034, +.054], paired exact $p = .385$, and low CORRUPT follow $13/321=.040$ [95% CI .024, .068]. The caveat is that a whole-prefix SEMANTIC-CF rewrite flips answers more often than single-token corruption, as expected for a broad edit, so it is not a gentle control; even so, the targeted numeric CORRUPT intervention still fails both load-bearing signatures, with the full SEMANTIC-CF numbers in the supplementary material. No single corruption statistic carries the conclusion: the whole battery does.

A last concern is that the force-continue probe is a two-pass re-prompt (a new prompt with the

Table 4: Selection flow and answer-eval strata. Answer-eval strata are stable-correct / SFT-gain / SFT-loss / both-wrong on the paired $n=400$ accuracy run. Probe-kept strata are the same case IDs after the separate CoT-generation eligibility filter. “No usable corrupt” is raw cases without a usable CORRUPT readout, including non-eligible cases and any eligible CoT with no corruptible numeric token. The final columns report present-condition CORRUPT follow rates in the two strata most relevant to the main claim. Full masked rows, loss/wrong strata, and sensitivity bounds are in `faithfulness_stats.md`.

Dataset/model	answer-eval strata	probe-kept strata	SFT correct	probe eligible	no usable corrupt	stable follow	gain follow
ChartQA 8B	265/46/20/69	262/39/10/10	311	321	79	9/262 (.034)	0/39 (.000)
ChartQA 32B	281/37/9/73	281/37/0/0	318	318	84	5/281 (.018)	0/37 (.000)
TabMWP 8B	345/40/1/14	345/40/0/0	385	385	15	3/345 (.009)	0/40 (.000)
TabMWP 32B	332/57/2/9	332/57/0/0	389	389	11	6/332 (.018)	0/57 (.000)

added instruction “Given ONLY the reasoning above, state ANSWER:”), so low present-image follow could in principle reflect the model re-reading the image on a second forward pass rather than a non-load-bearing chain. We address it with an IN-PLACE paradigm that stays inside the model’s own autoregressive stream: we re-feed the model’s *own* greedy tokens up to the boundary before a selected numeric token, substitute the corrupted value, and let the model *continue generating* its own chain and answer, with no re-prompt, no added instruction, and the conclusion line never supplied. A paradigm-validity control that re-feeds the *true* value and continues reproduces the model’s base answer on 287/307 cases (.935), certifying that the re-encoded continuation tracks the original stream, and the convergence threshold was pre-registered before the run (§8). Because this arm uses a ChartQA 8B student retrained for the in-place experiment, its absolute rates are not comparable to Table 2; the contrast is two-pass versus IN-PLACE *within* this one student. The stricter probe does raise the raw flip rate, with IN-PLACE CORRUPT flipping on .293 [95% CI .245,.346] of cases against .098 for the matched two-pass probe on the same student, and follow rising to .094 [95% CI .067,.132] from .036, so the two-pass paradigm *under*-estimates the raw flip rate. By the two-signature criterion (§3) the chain is still not load-bearing: follow stays low, snap still dominates at .733 [95% CI .681,.779], and most of the extra movement goes into “other” (53/307, .173). Stream-level corruption derails generation more than re-prompting, but rarely makes the model adopt the injected value as its answer. A low-powered no-visual variant (image absent throughout) agrees with the masked force-continue finding and is reported, with the full cross-paradigm table, in the supplementary material (`faithfulness_stats.md`).

6 Result III: Task-Dependent Information Sources and a Constructive Test

The chart/table probe (§5) isolates the *pixels* regime, where the answer reads from fixed visual evidence. The two substrates below are complementary diagnostics, not the same SFT-mechanism evidence restated: each tests a *different* readable source. Natural-image counting is an external regime check on *base* models that probes the *redundant-text* source, and the FinQA curriculum probes the *conclusion-line* source. We keep them separate rather than aggregating them into one causal claim.

Natural-image counting (external regime check; redundant-text source). To test whether the chart/table result is a chart artifact, we run a pre-specified TallyQA-complex counting probe on *base* Qwen3-VL-8B/32B, with no chart adapter. We treat this as an external regime check on the base model, not as SFT-internalization evidence on par with the ChartQA/TabMWP cells: there is no

distillation step here, so it tests only whether the non-load-bearing readout generalizes to a different visual regime, not whether an SFT gain was internalized. This is a genuinely perception-bottlenecked natural-image regime (free-form accuracy .415/.448). Yet the emitted CoT’s numeric content is again not load-bearing: across scales, follow is 0/345 with the image present and 1/345 when masked. Even when CORRUPT hits the count token itself, follow is 0/103 present and 1/103 masked. The model reconstructs the answer from redundant enumeration in the text, so masking the image is almost inert. This is the same reading-not-reasoning pattern with a different readable source: not the image, but the redundant chain.

Shortcut-removing curriculum (SFT; conclusion-line source). As a constructive counter-test, we train on rendered FinQA tables filtered to require multi-step, multi-cell arithmetic with a gold-program-verified flippable operand. The curriculum uses a deterministic gold-program teacher and a matched vanilla control; a text-teacher arm repeats the test with fluent answer-conditioned rationales. Accuracy rises substantially (8B: .080 base $\rightarrow$.582 vanilla $\rightarrow$.674 curriculum; 32B: .123 $\rightarrow$.659 $\rightarrow$.682). This probe yields *two distinct* readings that we deliberately do not merge. First, *operand-follow is near zero*: changing a verified on-path *input* operand yields follow 0/172 for the 8B curriculum and 0/175 for the 32B curriculum, so the emitted operands are not the load-bearing source. Second, *conclusion-line follow is high*: when the chain’s final conclusion is consistently rewritten to the recomputed value, the model follows it in 96–97% of cases, so the answer reads from the conclusion line. The observable mechanism is conclusion copying, not recomputation from the emitted operands under this probe; we do *not* extend this to the broader claim that no recomputation occurs, since the model may perform arithmetic latently and this probe cannot see it. We establish only that the written operands are not the load-bearing source. Even a shortcut-removing curriculum can therefore improve accuracy without installing a load-bearing written chain.

7 Discussion and Limitations

Implications. (i) Claims of “internalizing agentic reasoning” need causal audits. Significant accuracy gains are compatible with recovering the answer from still-visible evidence, reconstructing from redundant rationale text, or copying the rationale’s conclusion. (ii) The load-bearing locus is task-dependent: answer-time evidence can live in pixels, in redundant text, or in a conclusion line. A single CoT faithfulness score without information-source controls is under-specified.

Deployment audit implications. The probe is meant to be reused, not only to criticize. We read it as a small audit procedure for teams that ship CoT-style explanations on chart/table/figure QA. Before an emitted rationale is trusted, a team could run the targeted CORRUPT-vs.-no-target battery on a held-out slice and report F and the snap/follow split rather than accuracy alone. A CoT could then be surfaced to end users when follow is non-trivial and F is positive, i.e. when the written chain is load-bearing; when follow is near zero and the answer snaps back, the chain can instead be suppressed or labelled as a post-hoc summary rather than the decision path. A team that wants a release gate could set a faithfulness threshold per deployment, for example a minimum load-bearing F whose CI excludes zero, and treat regressions below it as a signal to revisit before shipping, alongside a short risk-disclosure note stating which fixed source the answer reads from (pixels, redundant text, or the conclusion line). Two misuse directions are symmetric and both wrong: an unfaithful CoT is *not* a license to withhold explanations entirely, and a fluent but non-load-bearing CoT should *not* be displayed to manufacture a false sense of transparency. The intended users are benchmark authors who report CoT internalization gains, VLM system

builders who decide whether to surface rationales, XAI auditors who certify explanation faithfulness, and operators of high-stakes table/chart QA (finance, scientific, and clinical reporting) where a plausible-but-decorative chain is an active hazard. In this framing the contribution is not a critique of CoT SFT claims but a reusable causal-audit practice.

Human trust and disclosure. The audit also has a human-centered reading. A fluent step-by-step rationale raises a reader’s perceived reliability of an answer, yet our probes show that this same rationale is often *not* the decision path: with the image present the model snaps back to the visible source while following the corrupted written intermediate near zero (e.g. cross-family ChartQA snap .990, follow .000), the FinQA curriculum copies the conclusion line (96–97%) while ignoring the emitted operands (0/172), and natural-image counting reconstructs the count from redundant chain text (0/345). Displaying such a chain manufactures *explanation-induced overtrust*: the explanation looks like a checkable arithmetic trace, but editing its numbers does not change the answer, so a user who audits the chain is auditing a post-hoc narrative rather than the computation. The risk is concrete in three settings. In *financial table QA*, a fluent “= revenue – cost” chain invites an analyst to trust a margin that, by our FinQA result, is read from a conclusion line the model can state without recomputing from the cited cells. In *education chart analysis*, a student shown a plausible reading of a bar chart may treat the rationale as a worked example, while the answer is recovered directly from pixels and the written steps are decorative. In *scientific figure QA*, a reviewer may over-credit a model that “explains” a value extracted from a plot whose intermediate quantities, if corrupted, leave the output unchanged. The disclosure rule follows from *which* source is load-bearing: when the answer reads from fixed visual evidence or a single conclusion line, the honest interface shows *structured evidence*, namely the highlighted cells, bars, or extracted operands the answer actually depends on together with the recomputed quantity, rather than a free-form CoT; a free-form chain should be surfaced as the decision rationale only when the probe shows it is load-bearing (positive F , non-trivial follow). This keeps the human-facing claim aligned with the causal evidence in this paper rather than with the chain’s fluency.

Limitations. All causal claims are about emitted rationales, not a proof that the model lacks latent internal computation. The re-reading step we infer from the snap behavior may itself involve latent arithmetic over the visible evidence, and our probes do not exclude that. What they do establish is narrower and specific: the final answer is not read off the corrupted written intermediate, since editing that token does not move the answer beyond no-target controls. The chart/table and FinQA operands are input-readable by design, which is exactly why these substrates can distinguish re-reading from written-chain use; the scope does not include tasks whose required operands are unavailable in the input. The 32B students use QLoRA/NF4 rather than full bf16 fine-tuning. The main teacher family is Qwen3-VL, although the FinQA text-teacher arm reduces a format-specific teacher concern. Corrupting one numeric token is a sensitivity lower bound; we therefore report it only with shuffle/paraphrase/filler/truncation, targeted local numeric and semantic-counterfactual controls, image masking, snap/follow/other classification, and targeted gold-program operand tests. The LOCAL-NUM control is currently a ChartQA 8B replication rather than a full eight-cell rerun; the whole-prefix SEMANTIC-CF diagnostic (supplementary material) uses a protected-final rewrite cache and is best read as a semantic diagnostic rather than a format-matched local control. Legacy batteries did not persist raw generated CoTs for non-eligible cases, so we cannot retrospectively split no-CoT, wrong-answer, and parse-failure reasons inside the nonprobe mass; instead, Table 4 reports the exact answer-eval and probe-kept strata and `faithfulness_stats.md` gives lower/upper follow bounds that assign the whole unobserved mass adversarially.

8 Conclusion

We set out to compress tool-scaffolded visual reasoning into one forward pass and to verify what moved. The answer is not a blanket dismissal of CoT or of distillation: SFT works, often strongly. The causal result is more specific. In these chart, table, natural-counting, and rendered-FinQA regimes, the emitted answer reads from the easiest available source: pixels, redundant enumerations, or the chain’s final conclusion. Accuracy can move while load-bearing written-chain faithfulness does not. That is the audit lesson behind *Reading, Not Reasoning*.

Reproducibility Statement

All numbers regenerate from a fingerprinted, append-only result store (`results.jsonl`) via the released scripts; SFT, merge, battery, curriculum, and probe scripts are included. ChartQA/TabMWP builders, power table, faithfulness figures, N3 natural-counting probe, and N1 targeted curriculum probe each correspond to one script with fixed seeds or saved artifacts. The generation-time IN-PLACE paradigm (`battery_online_p3.py`) and its convergence threshold were pre-registered before the run in `docs/preregistration_p3_inplace.md`; both paradigms in the cross-paradigm comparison run on the same student so the only varied factor is prompt-level versus stream-level intervention.

A Artifact Map

Table 5 gives the minimum artifact map needed to reconstruct each headline table or figure without reading source code first. Paths are relative to the repository root. JSON hashes are 12-character SHA-256 prefixes of the files used in this draft.

References

- Ashtosh Adhikari and Mirella Lapata. Multimodal latent reasoning via predictive embeddings. *arXiv preprint arXiv:2604.08065*, 2026.
- Shuai Bai et al. Qwen3-VL technical report. *arXiv preprint arXiv:2511.21631*, 2025.
- Oliver Bentham, Nathan Stringham, and Ana Marasovic. Chain-of-thought unfaithfulness as disguised accuracy. *Transactions on Machine Learning Research*, 2024.
- Victor Carbune, Hassan Mansoor, Fangyu Liu, Rahul Aralikkatte, Gilles Baechler, Jindong Chen, and Abhanshu Sharma. Chart-based reasoning: Transferring capabilities from LLMs to VLMs. *arXiv preprint arXiv:2403.12596*, 2024.
- Lei Chen, Xuanle Zhao, Zhixiong Zeng, Jing Huang, Yufeng Zhong, and Lin Ma. Chart-R1: Chain-of-thought supervision and reinforcement for advanced chart reasoner. *arXiv preprint arXiv:2507.15509*, 2025.
- Wenhu Chen, Xueguang Ma, Xinyi Wang, and William W. Cohen. Program of thoughts prompting: Disentangling computation from reasoning for numerical reasoning tasks. *Transactions on Machine Learning Research*, 2023.
- Zhiyu Chen, Wenhu Chen, Charese Smiley, Sameena Shah, Iana Borova, Dylan Langdon, Reema Moussa, Matt Beane, Ting-Hao Huang, Bryan Routledge, and William Yang Wang. FinQA: A dataset of numerical reasoning over financial data. In *EMNLP*, 2021.

- Jay DeYoung, Sarthak Jain, Nazneen Fatema Rajani, Eric Lehman, Caiming Xiong, Richard Socher, and Byron C. Wallace. ERASER: A benchmark to evaluate rationalized NLP models. In *ACL*, 2020.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *ICML*, 2023.
- Xinhai Hou, Shaoyuan Xu, Manan Biyani, Moyan Li, Jia Liu, Todd C. Hollon, and Bryan Wang. CodeV: Code with images for faithful visual reasoning via tool-aware policy optimization. *arXiv preprint arXiv:2511.19661*, 2026. CVPR 2026 Oral.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of ACL*, 2023.
- Yushi Hu, Otilia Stretcu, Chun-Ta Lu, Krishnamurthy Viswanathan, Kenji Hata, Enming Luo, Ranjay Krishna, and Ariel Fuxman. Visual program distillation: Distilling tools and programmatic reasoning into vision-language models. In *CVPR*, 2024.
- Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *ACL*, 2020.
- Kushal Kafle, Brian Price, Scott Cohen, and Christopher Kanan. DVQA: Understanding data visualizations via question answering. In *CVPR*, 2018.
- Samira Ebrahimi Kahou, Vincent Michalski, Adam Atkinson, Ákos Kádár, Adam Trischler, and Yoshua Bengio. FigureQA: An annotated figure dataset for visual reasoning. *ICLR Workshop*, 2018.
- Takeshi Kojima, Shixiang Shane Gu, Machel Reid, Yutaka Matsuo, and Yusuke Iwasawa. Large language models are zero-shot reasoners. In *Advances in Neural Information Processing Systems*, 2022.
- Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, Esin Durmus, Evan Hubinger, Jackson Kernion, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.
- You Li, Chi Chen, Yanghao Li, Fanhu Zeng, Kaiyu Huang, Jinan Xu, and Maosong Sun. Imagination helps visual reasoning, but not yet in latent space. *arXiv preprint arXiv:2602.22766*, 2026. ICML 2026.
- Junteng Liu, Weihao Zeng, Xiwen Zhang, Yijun Wang, Zifei Shan, and Junxian He. On the perception bottleneck of VLMs for chart understanding. *arXiv preprint arXiv:2503.18435*, 2025a.
- Zujing Liu, Junwen Pan, Qi She, Yuan Gao, and Guisong Xia. On the faithfulness of visual thinking: Measurement and enhancement. *arXiv preprint arXiv:2510.23482*, 2025b.
- Pan Lu, Liang Qiu, Kai-Wei Chang, Ying Nian Wu, Song-Chun Zhu, Tanmay Rajpurohit, Peter Clark, and Ashwin Kalyan. Dynamic prompt learning via policy gradient for semi-structured mathematical reasoning. In *ICLR*, 2023.

- Qing Lyu, Shreya Havaldar, Adam Stein, Li Zhang, Delip Rao, Eric Wong, Marianna Apidianaki, and Chris Callison-Burch. Faithful chain-of-thought reasoning. In *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 305–329, 2023.
- Ahmed Masry, Do Xuan Long, Jia Qing Tan, Shafiq Joty, and Enamul Hoque. ChartQA: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of ACL*, 2022.
- Nitesh Methani, Pritha Ganguly, Mitesh M. Khapra, and Pratyush Kumar. PlotQA: Reasoning over scientific plots. In *WACV*, 2020.
- Jacob Pfau, William Merrill, and Samuel R. Bowman. Let’s think dot by dot: Hidden computation in transformer language models. *arXiv preprint arXiv:2404.15758*, 2024.
- Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Eric Hambro, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. Toolformer: Language models can teach themselves to use tools. In *Advances in Neural Information Processing Systems*, 2023.
- Philip Wootack Shin, Jack Sampson, Vijaykrishnan Narayanan, Andres Marquez, and Mahantesh Halappanavar. Losing the plot: How VLM responses degrade on imperfect charts. *arXiv preprint arXiv:2509.18425*, 2025.
- Liyan Tang, Grace Kim, Xinyu Zhao, Thom Lake, Wenxuan Ding, Fangcong Yin, Prasann Singhal, Manya Wadhwa, Zeyu Leo Liu, Zayne Sprague, Ramya Namuduri, Bodun Hu, Juan Diego Rodriguez, Puyuan Peng, and Greg Durrett. ChartMuseum: Testing visual reasoning capabilities of large vision-language models. *arXiv preprint arXiv:2505.13444*, 2025.
- Zhou Tao, Fang Zhang, Zewen Ding, Shida Wang, Xiaokun Sun, YongXiang Hua, Haoyu Cao, and Linli Xu. LOCUS: Local visual cue search for enhancing fine-grained perception in multimodal large language models. *arXiv preprint arXiv:2606.16586*, 2026.
- Miles Turpin, Julian Michael, Ethan Perez, and Samuel R. Bowman. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting. *Advances in Neural Information Processing Systems*, 2023.
- Martin Tutek, Fateme Hashemi Chaleshtori, Ana Marasovic, and Yonatan Belinkov. Measuring chain of thought faithfulness by unlearning reasoning steps. *arXiv preprint arXiv:2502.14829*, 2025.
- Weiyun Wang, Zhangwei Gao, Lixin Gu, Hengjun Pu, Long Cui, Xingguang Wei, Zhaoyang Liu, Linglin Jing, Shenglong Ye, Jie Shao, et al. InternVL3.5: Advancing open-source multimodal models in versatility, reasoning, and efficiency. *arXiv preprint arXiv:2508.18265*, 2025.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *ICLR*, 2023.
- Zirui Wang, Mengzhou Xia, Luxi He, Howard Chen, Yitao Liu, Richard Zhu, Kaiqu Liang, Xindi Wu, Haotian Liu, Sadhika Malladi, Alexis Chevalier, Sanjeev Arora, and Danqi Chen. CharXiv: Charting gaps in realistic chart understanding in multimodal LLMs. *arXiv preprint arXiv:2406.18521*, 2024.

- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed H. Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems*, 2022.
- Lai Wei, Liangbo He, Jun Lan, Lingzhong Dong, Yutong Cai, Siyuan Li, Huijia Zhu, Weiqiang Wang, Linghe Kong, Yue Wang, Zhuosheng Zhang, and Weiran Huang. Zooming without zooming: Region-to-image distillation for fine-grained multimodal perception. *arXiv preprint arXiv:2602.11858*, 2026.
- Sarah Wiegrefe and Ana Marasović. Teach me to explain: A review of datasets for explainable natural language processing. In *NeurIPS Datasets and Benchmarks Track*, 2021.
- Fengbin Zhu, Wenqiang Lei, Youcheng Huang, Chao Wang, Shuo Zhang, Jiancheng Lv, Fuli Feng, and Tat-Seng Chua. TAT-QA: A question answering benchmark on a hybrid of tabular and textual content in finance. In *ACL*, 2021.

Table 5: Primary artifacts for the SFT and probe results. Probe edits use greedy decoding and fixed edit seed `Random(0)` unless otherwise noted; result-store rows also carry per-run fingerprints over split, method, decoding, seed, prompt, and code version.

Result family	Result JSON / store	Script and seed	Checkpoint or input path
Global result store	data/distill/results/results.jsonl, SHA 31ef62ccb77a	scripts/regen_tables.py; stored run seed/fingerprint	append-only store used by tables and map figures
ChartQA LoRA SFT accuracy	data/distill/poc/lora_8b_chartqa/eval_n400.json, SHA 2ce4d9175235; data/distill/poc/lora_32b_chartqa/eval_n400.json, SHA 92e59e633a71	scripts/poc_sft.py; scripts/poc_sft_32b_qlora.py; scripts/eval_sft_n400.py	train data/distill/poc/chartqa_cot_train.jsonl; adapters data/distill/poc/lora_8b_chartqa, data/distill/poc/lora_32b_chartqa/epoch_1
TabMWP LoRA SFT accuracy	data/distill/poc/eval_sft_8b_tabmwp_n400.json, SHA 95aaba7b47df; data/distill/poc/eval_sft_32b_tabmwp_n400.json, SHA b08953be8f1f	scripts/run_wu3_stage2.sh; scripts/eval_sft_n400.py	train data/distill/poc/tabmwp_cot_train.jsonl; adapters data/distill/poc/lora_8b_tabmwp, data/distill/poc/lora_32b_tabmwp/epoch_4
ChartQA faithfulness battery	data/distill/poc/battery_8b_present.json, SHA 0ff0da834fc9; data/distill/poc/battery_32b_present.json, SHA 34a57e280224; masked companions in same directory	scripts/battery_n400.py, edit seed 0	same ChartQA adapters; test split data/distill/chartqa/test_cases_400.jsonl
Local numeric control	data/distill/poc/battery_p0_local_chartqa8b_present.json, SHA e8f8867af47f	scripts/battery_n400.py, SHA 63b51bf0af0a; interventions corrupt local_num, edit seed 0	ChartQA 8B adapter; test split data/distill/chartqa/test_cases_400.jsonl
Semantic counterfactual diagnostic	data/distill/poc/battery_p0_semantic_chartqa8b_present.json, SHA 9784029ee5a2; rewrite cache data/distill/poc/semantic_cf_p0_chartqa8b_present.jsonl, SHA 1d54f6ee7890	scripts/battery_n400.py, SHA 63b51bf0af0a; interventions corrupt semantic_cf, edit seed 0; ORCH rewrite endpoint with explicit ChartQA 8B consent	ChartQA 8B adapter; test split data/distill/chartqa/test_cases_400.jsonl
Cross-family replication	data/distill/poc/battery_p0_cross_family_internvl35_chartqa_present.json, SHA 3403bd84b51d; base/probe caches data/distill/poc/internvl35_chartqa_base_cot.jsonl, SHA 017cdeedd221; data/distill/poc/internvl35_chartqa_probe.jsonl, SHA 1ddb3f08b58	scripts/battery_cross_family_endpoint.py, SHA cbeb6e748946; interventions corrupt shuffle, edit seed 0; endpoint greedy decode	InternVL3.5-8B served locally; ChartQA test split data/distill/chartqa/test_cases_400.jsonl
Generation-time (in-place) probe	data/distill/poc/battery_p3_online_chartqa8b_present.json, SHA e21d22b46ff9; no-visual companion data/distill/poc/battery_p3_online_chartqa8b_maskedB.json, SHA d8fce3d310e2; matched two-pass data/distill/poc/battery_p3_twopass_chartqa8b_present.json, SHA 083ccf248e8b	scripts/battery_online_p3.py, SHA b82ac920830c, greedy, edit seed 0; pre-registration docs/preregistration_p3_inplace.md	retrained ChartQA 8B adapter data/distill/poc/lora_8b_chartqa; test split data/distill/chartqa/test_cases_400.jsonl
TabMWP faithfulness battery	data/distill/poc/battery_tabmwp8b_present.json, SHA c27d11005883; data/distill/poc/battery_tabmwp32b_present.json, SHA 078a37c4e4c6; masked companions in same directory	scripts/battery_n400.py, edit seed 0	same TabMWP adapters; test split data/distill/tabmwp/test_cases_400.jsonl
Faithfulness statistical and selection audit	data/distill/results/faithfulness_stats.json, SHA 28743bfa46e7; Markdown summary data/distill/results/faithfulness_stats.md, SHA fd789992538a	scripts/faithfulness_stats.py, SHA 5f4318e19dc8; CPU-only fixed bootstrap seeds	consumes ChartQA/TabMWP battery files, local/semantic-control and cross-family battery files when present, SFT prediction JSONL files, and FinQA targeted battery files; emits raw-to-probe flow, answer-eval strata, gain-subset, local/semantic/cross-family tests, and sensitivity bounds
FinQA targeted curriculum probe	data/distill/poc/battery_n1_8b.json, SHA 03dfe86ad4e; data/distill/poc/battery_n1_32b.json, SHA c808008b06b9; text-teacher and dense-control companions in same directory	scripts/build_n1_curriculum.py; scripts/run_n1_sft.sh; scripts/battery_n1_targeted.py, edit seed 0	train data/distill/finqa/curriculum_dev.*; probe split data/distill/finqa/curriculum_test_strict.jsonl; LoRA and QLoRA adapters under data/distill/poc/lora.*_finqa.*
Dense 8B control	data/distill/poc/full_8b_chartqa/full_sft_summary.json; data/distill/poc/full_8b_tabmwp/full_sft_summary.json; dense battery files under data/distill/poc/battery_full8b.*	scripts/audit_full_sft_8b_nonvideo.py; scripts/eval_full_sft_n400.py; scripts/export_full_sft_8b_nonvideo_evidence.py	full-SFT checkpoints under data/distill/poc/full_8b.*; summaries record trainable fraction and freeze policy